

Mathematical English

Physics

English Vocabulary, Japanese Meaning,
Mathematical Formulas, and Physical Meaning

Contents

1	Classical Mechanics	3
1.1	Basic Concepts	3
1.2	Kinematics	3
1.3	Newton's Laws	3
1.4	Momentum Conservation	4
2	Rotational Motion	4
3	Gravitation	5
3.1	Newton's Law of Universal Gravitation	5
4	Oscillations	5
4.1	Simple Harmonic Motion	5
4.2	Mass-Spring System	5
4.3	Simple Pendulum	6
5	Waves	6
6	Fluid Mechanics	6
6.1	Density	6
6.2	Pressure	7
6.3	Continuity Equation	7
6.4	Bernoulli's Equation	7
7	Thermodynamics	7
7.1	First Law of Thermodynamics	7
7.2	Ideal Gas Law	8
7.3	Entropy	8
8	Electromagnetism	8
8.1	Electric Charge	8
8.2	Coulomb's Law	8
8.3	Electric Field	8
8.4	Electric Potential	8
8.5	Ohm's Law	9
9	Maxwell's Equations	9
9.1	Gauss's Law for Electricity	9
9.2	Gauss's Law for Magnetism	9
9.3	Faraday's Law	9
9.4	Ampère–Maxwell Law	9
10	Special Relativity	9
10.1	Lorentz Factor	9
10.2	Time Dilation	10
10.3	Length Contraction	10
10.4	Mass-Energy Equivalence	10

11 Quantum Mechanics	10
11.1 Planck Relation	10
11.2 de Broglie Wavelength	10
11.3 Heisenberg Uncertainty Principle	10
11.4 Schrödinger Equation	10
11.5 Probability Density	11
12 Statistical Mechanics	11
12.1 Boltzmann Distribution	11
12.2 Boltzmann Entropy	11
12.3 Partition Function	11
13 Lagrangian and Hamiltonian Mechanics	11
13.1 Lagrangian	11
13.2 Hamiltonian	12
14 Core Physical Constants	12
15 Common Physics Symbols	12
16 Key Distinctions	13
16.1 Mass vs. Weight	13
16.2 Energy vs. Power	13
16.3 Velocity vs. Acceleration	13
17 Physics Learning Map	13
18 Core Relationships	14
19 Physics and Engineering	14

1 Classical Mechanics

1.1 Basic Concepts

English	Formula	Meaning
Position	$x(t)$	The location of an object as a function of time.
Displacement	$\Delta x = x_f - x_i$	Change in position.
Velocity	$v = \frac{dx}{dt}$	Rate of change of position.
Acceleration	$a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$	Rate of change of velocity.
Mass	m	Measure of an object's inertia.
Momentum	$\mathbf{p} = m\mathbf{v}$	Quantity of motion carried by an object.
Force	$\mathbf{F} = m\mathbf{a}$	Interaction that changes an object's motion.
Work	$W = \int \mathbf{F} \cdot d\mathbf{r}$	Energy transferred by a force acting through a displacement.
Kinetic Energy	$K = \frac{1}{2}mv^2$	Energy associated with motion.
Potential Energy	$U(\mathbf{r})$	Energy associated with position or configuration.
Mechanical Energy	$E = K + U$	Sum of kinetic and potential energy.
Power	$P = \frac{dW}{dt}$	Rate at which work is performed.

1.2 Kinematics

For constant acceleration,

$$v = v_0 + at$$

$$x = x_0 + v_0t + \frac{1}{2}at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

These equations describe motion without requiring an explicit analysis of the forces causing the motion.

1.3 Newton's Laws

Newton's First Law

An object remains at rest or moves with constant velocity unless acted upon by a net external force.

$$\sum \mathbf{F} = 0$$

Newton's Second Law

$$\sum \mathbf{F} = m\mathbf{a}$$

The net force determines the acceleration of an object.

Newton's Third Law

$$\mathbf{F}_{AB} = -\mathbf{F}_{BA}$$

For every force exerted by one object on another, there is an equal and opposite force.

1.4 Momentum Conservation

For an isolated system,

$$\mathbf{p}_{\text{initial}} = \mathbf{p}_{\text{final}}$$

or

$$\sum_i m_i \mathbf{v}_i = \text{constant.}$$

Momentum conservation is particularly important in collision and interaction problems.

2 Rotational Motion

English	Formula	Meaning
Angular Position	θ	Position measured by an angle.
Angular Velocity	$\omega = \frac{d\theta}{dt}$	Rate of change of angular position.
Angular Acceleration	$\alpha = \frac{d\omega}{dt}$	Rate of change of angular velocity.
Torque	$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$	Rotational effect of a force.
Moment of Inertia	$I = \sum_i m_i r_i^2$	Rotational analogue of mass.
Angular Momentum	$\mathbf{L} = \mathbf{r} \times \mathbf{p}$	Rotational analogue of linear momentum.
Rotational Energy	$K_{\text{rot}} = \frac{1}{2} I \omega^2$	Kinetic energy associated with rotation.

The rotational form of Newton's second law is

$$\boldsymbol{\tau} = I\boldsymbol{\alpha}.$$

For a rigid body rotating about a fixed axis,

$$L = I\omega.$$

If the net external torque is zero,

$$L = \text{constant.}$$

3 Gravitation

3.1 Newton's Law of Universal Gravitation

$$F = G \frac{m_1 m_2}{r^2}$$

where

$$G \approx 6.674 \times 10^{-11} \text{ N m}^2/\text{kg}^2.$$

English: gravitational force

:

Meaning: The gravitational force between two masses decreases with the square of the distance between them.

Near the Earth's surface,

$$F_g = mg.$$

Thus,

$$g \approx 9.81 \text{ m/s}^2.$$

4 Oscillations

4.1 Simple Harmonic Motion

Simple harmonic motion is described by

$$x(t) = A \cos(\omega t + \phi).$$

The velocity is

$$v(t) = -A\omega \sin(\omega t + \phi),$$

and the acceleration is

$$a(t) = -A\omega^2 \cos(\omega t + \phi).$$

Therefore,

$$a = -\omega^2 x.$$

This relationship is the defining characteristic of simple harmonic motion.

4.2 Mass-Spring System

For a spring,

$$F = -kx.$$

The angular frequency is

$$\omega = \sqrt{\frac{k}{m}}.$$

The period is

$$T = 2\pi\sqrt{\frac{m}{k}}.$$

The total mechanical energy is

$$E = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 = \frac{1}{2}kA^2.$$

4.3 Simple Pendulum

For small oscillations,

$$T = 2\pi\sqrt{\frac{L}{g}}.$$

The small-angle approximation is

$$\sin \theta \approx \theta.$$

5 Waves

English	Formula	Meaning
Wavelength	λ	Distance between corresponding points of consecutive waves.
Frequency	f	Number of cycles per unit time.
Period	$T = \frac{1}{f}$	Time required for one complete cycle.
Wave Speed	$v = f\lambda$	Speed at which the wave propagates.
Amplitude	A	Maximum displacement from equilibrium.
Wave Number	$k = \frac{2\pi}{\lambda}$	Spatial frequency of a wave.
Angular Frequency	$\omega = 2\pi f$	Frequency expressed in radians per second.

A one-dimensional traveling wave can be written as

$$y(x, t) = A \cos(kx - \omega t + \phi).$$

6 Fluid Mechanics

6.1 Density

$$\rho = \frac{m}{V}.$$

Density =

Density represents mass per unit volume.

6.2 Pressure

$$P = \frac{F}{A}.$$

Pressure is force distributed over an area.

For a fluid at rest,

$$P = P_0 + \rho gh.$$

6.3 Continuity Equation

For an incompressible steady flow,

$$A_1 v_1 = A_2 v_2.$$

This expresses conservation of mass.

6.4 Bernoulli's Equation

For an ideal incompressible steady flow,

$$P + \frac{1}{2} \rho v^2 + \rho gh = \text{constant}.$$

The three terms represent pressure energy, kinetic energy per unit volume, and gravitational potential energy per unit volume.

7 Thermodynamics

English	Formula	Meaning
Temperature	T	Measure related to thermal energy and equilibrium.
Heat	Q	Energy transferred because of a temperature difference.
Internal Energy	U	Energy contained within a system at the microscopic level.
Entropy	S	State function associated with energy dispersal and the number of accessible microscopic states.
Enthalpy	$H = U + PV$	Useful thermodynamic potential, especially for constant-pressure processes.

7.1 First Law of Thermodynamics

$$\Delta U = Q - W.$$

Here, W is work done by the system.

This equation expresses conservation of energy in thermodynamic systems.

7.2 Ideal Gas Law

$$PV = nRT.$$

where

$$R = 8.314 \text{ J}/(\text{mol K}).$$

7.3 Entropy

For a reversible process,

$$dS = \frac{\delta Q_{\text{rev}}}{T}.$$

For an isolated system,

$$\Delta S \geq 0.$$

This is one mathematical expression of the second law of thermodynamics.

8 Electromagnetism

8.1 Electric Charge

Electric charge =

The elementary charge is

$$e \approx 1.602 \times 10^{-19} \text{ C}.$$

8.2 Coulomb's Law

$$F = k_e \frac{|q_1 q_2|}{r^2}.$$

In vector form,

$$\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{\mathbf{r}}.$$

8.3 Electric Field

$$\mathbf{E} = \frac{\mathbf{F}}{q}.$$

For a point charge,

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}.$$

8.4 Electric Potential

$$V = \frac{U}{q}.$$

For a point charge,

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}.$$

The relationship between electric field and potential is

$$\mathbf{E} = -\nabla V.$$

8.5 Ohm's Law

$$V = IR.$$

where

- V = voltage
- I = electric current
- R = resistance

Electrical power is

$$P = VI = I^2 R = \frac{V^2}{R}.$$

9 Maxwell's Equations

Maxwell's equations provide a compact mathematical description of classical electromagnetism.

9.1 Gauss's Law for Electricity

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}.$$

9.2 Gauss's Law for Magnetism

$$\nabla \cdot \mathbf{B} = 0.$$

This expresses the absence of isolated magnetic monopoles in classical electromagnetism.

9.3 Faraday's Law

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$

A changing magnetic field produces an electric field.

9.4 Ampère–Maxwell Law

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

A current and a changing electric field produce a magnetic field.

10 Special Relativity

10.1 Lorentz Factor

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}.$$

where

$$c \approx 3.00 \times 10^8 \text{ m/s}.$$

10.2 Time Dilation

$$\Delta t = \gamma \Delta \tau.$$

Moving clocks are measured to run slower relative to an observer in another inertial frame.

10.3 Length Contraction

$$L = \frac{L_0}{\gamma}.$$

10.4 Mass-Energy Equivalence

$$E = mc^2.$$

More generally, relativistic energy and momentum satisfy

$$E^2 = p^2 c^2 + m^2 c^4.$$

11 Quantum Mechanics

11.1 Planck Relation

$$E = hf = \hbar \omega.$$

where

$$\hbar = \frac{h}{2\pi}.$$

11.2 de Broglie Wavelength

$$\lambda = \frac{h}{p}.$$

Particles can exhibit wave-like behavior.

11.3 Heisenberg Uncertainty Principle

$$\Delta x \Delta p \geq \frac{\hbar}{2}.$$

Position and momentum cannot simultaneously have arbitrarily precise values.

11.4 Schrödinger Equation

The time-dependent Schrödinger equation is

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H} \psi.$$

For a non-relativistic particle in a potential V ,

$$i\hbar \frac{\partial \psi}{\partial t} = \left[-\frac{\hbar^2}{2m} \nabla^2 + V \right] \psi.$$

The time-independent equation is

$$\hat{H} \psi = E \psi.$$

11.5 Probability Density

The probability density is

$$P(x) = |\psi(x)|^2.$$

The wave function must satisfy the normalization condition

$$\int |\psi(x)|^2 dx = 1.$$

12 Statistical Mechanics

12.1 Boltzmann Distribution

$$P(E) \propto e^{-E/(k_B T)}.$$

where k_B is the Boltzmann constant.

12.2 Boltzmann Entropy

$$S = k_B \ln \Omega.$$

Here, Ω represents the number of accessible microscopic states.

12.3 Partition Function

$$Z = \sum_i e^{-\beta E_i},$$

where

$$\beta = \frac{1}{k_B T}.$$

The partition function is a central quantity in statistical mechanics.

13 Lagrangian and Hamiltonian Mechanics

13.1 Lagrangian

The Lagrangian is defined as

$$L = T - U.$$

The equations of motion follow from the Euler–Lagrange equation:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0.$$

This formulation can describe classical mechanics without directly resolving every force component.

13.2 Hamiltonian

The Hamiltonian is

$$H = \sum_i p_i \dot{q}_i - L.$$

Hamilton's equations are

$$\dot{q}_i = \frac{\partial H}{\partial p_i},$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}.$$

In many physical systems, the Hamiltonian corresponds to the total energy.

14 Core Physical Constants

Constant	Symbol	Approximate Value
Speed of light	c	3.00×10^8 m/s
Gravitational constant	G	6.674×10^{-11} N m ² /kg ²
Planck constant	h	6.626×10^{-34} J s
Reduced Planck constant	$\hbar$	1.055×10^{-34} J s
Boltzmann constant	k_B	1.381×10^{-23} J/K
Elementary charge	e	1.602×10^{-19} C
Vacuum permittivity	ε_0	8.854×10^{-12} F/m
Vacuum permeability	μ_0	approximately 1.257×10^{-6} H/m

15 Common Physics Symbols

Symbol	English
x	position
v	velocity
a	acceleration
m	mass
$\mathbf{p}$	momentum
$\mathbf{F}$	force
E	energy
K	kinetic energy
U	potential energy
P	power / pressure
ω	angular frequency
λ	wavelength
f	frequency
θ	angle
ρ	density / charge density
ψ	wave function
H	Hamiltonian

16 Key Distinctions

16.1 Mass vs. Weight

Mass is a measure of inertia:

$$m.$$

Weight is a gravitational force:

$$W = mg.$$

Mass is measured in kilograms, while weight is measured in newtons.

16.2 Energy vs. Power

Energy measures the amount of energy transferred or stored:

$$E.$$

Power measures the rate of energy transfer:

$$P = \frac{dE}{dt}.$$

16.3 Velocity vs. Acceleration

Velocity describes the rate of change of position:

$$v = \frac{dx}{dt}.$$

Acceleration describes the rate of change of velocity:

$$a = \frac{dv}{dt}.$$

An object can have nonzero velocity while having zero acceleration.

17 Physics Learning Map

A useful conceptual progression is:

Mechanics → Waves → Thermodynamics → Electromagnetism → Relativity → Quantum Mechanics

Mathematics provides the language for describing these physical theories.

A more mathematical progression is:

Algebra → Calculus → Differential Equations → Linear Algebra → Vector Calculus → Advanced Physics.

18 Core Relationships

Some of the most important relationships in introductory physics are:

$$F = ma$$

$$p = mv$$

$$E = K + U$$

$$W = \Delta E$$

$$E = mc^2$$

$$E = hf$$

$$\lambda = \frac{h}{p}$$

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$$

These equations illustrate how physical concepts are expressed mathematically.

19 Physics and Engineering

Physics provides fundamental models of nature, while engineering applies these models to designed systems.

For example,

$$F = ma$$

can be used in mechanical system design,

$$V = IR$$

in electrical systems,

$$PV = nRT$$

in thermodynamic systems, and

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$$

in quantum systems.

For computational and engineering applications, physical models are often combined with numerical methods, optimization, simulation, and data analysis.